# Particle Physics 3 Supersymmetry And Grand Unification: Supersymmetry & Grand Unification: Lecture 8

Leonard Susskind

Transcript-derived notes curated by LazyingArt LLC

Original lecture by Leonard Susskind. Transcript-derived notes curated by LazyingArt LLC.

# Chapter 1

# Supersymmetry in Four Dimensions

The previous lecture treated supersymmetry in a toy world with only time. Leonard Susskind's Lecture 8 begins by undoing that simplification. If energy is part of a four-vector, then a supersymmetry algebra involving only the Hamiltonian cannot be the full relativistic story. The lecture then does two things in sequence. First, it builds the four-dimensional algebra carefully enough that we see why spinorial Grassmann variables are unavoidable. Second, it constructs a simple supersymmetric field theory and follows the bookkeeping until ordinary bosons, fermions, masses, vertices, and cancellations appear. The present chapter is curated from the lecture by LazyingArt LLC, and it keeps the lecture's order on purpose: the formalism is introduced only when the preceding difficulty has made it necessary.

## 1.1 From Time-Only Supersymmetry to Lorentz Invariance

The starting point is the one-dimensional supersymmetry algebra from the previous lecture,

$$\{Q^\dagger, Q\} = 2H, \tag{1.1}$$

where the right-hand side involves only the Hamiltonian. In a purely time-dependent model this is natural. But now the lecture asks the objection that forces everything that follows: energy is only the time component of momentum, so how can a relativistic theory speak about $H$ and ignore the spatial components?

That question is not merely cosmetic. Under a Lorentz transformation, the time component of a four-vector mixes with the spatial ones. So if the algebra is truly Lorentz covariant, the right-hand side of the anticommutator cannot remain a lone $H$. It must know about the whole four-momentum $P_\mu$. The lecture therefore pivots at once from the toy model to the relativistic problem.

But instead of writing the finished superalgebra immediately, Susskind does something more instructive. He asks what kind of relativistic object ought to sit at the heart of the construction. To get a feel for that, he returns to the massless Dirac equation in its simplest two-component form.

## 1.2 Massless Chiral Fermions and the Hamiltonian $H = \sigma \cdot p$

A massive Dirac fermion comes with both handednesses. A massless fermion need not. This is the first physical fact the lecture wants to establish, because in four dimensions the supersymmetry

generators and the Grassmann coordinates themselves will have to transform as spinors, and the cleanest relativistic spinor on the table is the massless chiral one.

For such a particle we work with a two-component spinor,

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}, \qquad \text{or, with an index, } \psi_\alpha. \tag{1.2}$$

The lecture discusses helicity and chirality in deliberately practical terms. Helicity is tied to $\sigma \cdot p$, chirality to the gamma-matrix structure; in the massless case under discussion the two coincide, and that is the case we need here.

### 1.2.1 Question & Answer

**Question.** Why can a massless particle be purely one-handed, whereas a massive particle cannot?

**Answer.** For a massive particle, helicity is frame-dependent. One can always boost to a frame that overtakes the particle. Its spin does not flip, but its momentum reverses relative to the observer, so a state that looked right-handed can look left-handed. A massive relativistic theory must therefore contain both handednesses. A genuinely massless particle moves at the speed of light, so it cannot be overtaken. In the massless case, where chirality and helicity coincide, a purely one-handed description is consistent.

With that picture in mind, the lecture writes the Hamiltonian

$$H = \boldsymbol{\sigma} \cdot \mathbf{p}. \tag{1.3}$$

Here $\boldsymbol{\sigma} = (\sigma^1, \sigma^2, \sigma^3)$ are the Pauli matrices. Susskind immediately interprets the sign of $\sigma \cdot p$: the relative orientation of spin and momentum determines the sign of the energy, and one then proceeds to the usual relativistic reinterpretation of negative-energy states.

The next move is concrete. Replace $H$ by $i\partial_t$ and $p_i$ by $-i\partial_i$. Then the Hamiltonian equation becomes

$$i\frac{\partial \psi}{\partial t} = -\, i\, \sigma^i \partial_i \psi. \tag{1.4}$$

The lecture then rewrites this first in a partially covariant form,

$$i\frac{\partial \psi}{\partial x^0} + i\, \sigma^i \partial_i \psi = 0, \tag{1.5}$$

and only afterward introduces the final compact notation. Defining

$$\sigma^0 = \mathbf{1}, \tag{1.6}$$

we may combine the four pieces into

$$i\, \sigma^\mu \partial_\mu \psi = 0. \tag{1.7}$$

This is the chiral Dirac equation in the lecture's form. The important narrative point is that the equation was not introduced as abstract notation. It arose by starting with the chiral Hamiltonian $H = \sigma \cdot p$, turning it into a wave equation, and only then packaging it into a Lorentz-looking formula. That rhythm matters because it prepares the next leap: if the relativistic fermion is a two-component anticommuting object, then the Grassmann coordinates of superspace must be as well.

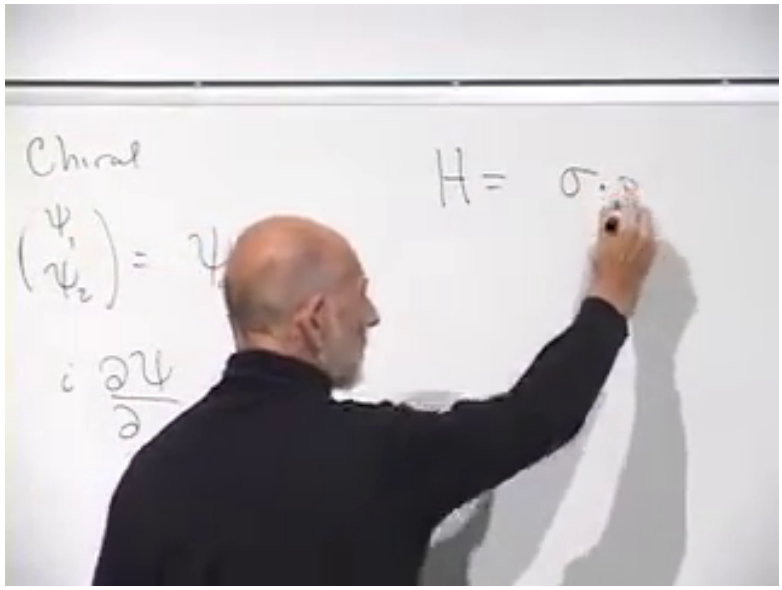


Figure 1.1: Chiral-spinor setup and the Hamiltonian $H = \sigma \cdot p$.

## 1.3 The Four-Dimensional Superalgebra

We now return to supersymmetry proper. The chiral spinor has shown us what kind of object Lorentz invariance wants. So the Grassmann coordinates cannot remain a single $\theta$; they must be $\theta_\alpha$ and their conjugates $\bar{\theta}_\alpha$. There are therefore four real Grassmann directions altogether. A general superfield is a function of

$$x^\mu, \qquad \theta_\alpha, \qquad \bar{\theta}_\alpha,$$

and because the Grassmann variables anticommute, its expansion terminates at fourth order. The lecture emphasizes this bookkeeping point before doing anything more sophisticated.

Now we ask what replaces the one-dimensional relation $\{Q^\dagger, Q\} = 2H$. The right-hand side must involve $P_\mu$, but that immediately creates an index mismatch. The left-hand side carries spinor indices $\alpha, \beta$; the momentum carries a Lorentz index $\mu$. The sigma matrices enter because something has to soak up $\mu$ and return the pair of spinor indices. This is one of the lecture's most characteristic moves: the next mathematical object is introduced because the previous formula quite visibly does not fit together.

In schematic lecture form, the four-dimensional algebra is

$$\{Q^\dagger_\alpha, Q_\beta\} = 2\,\sigma^\mu_{\alpha\beta} P_\mu = 2i\,\sigma^\mu_{\alpha\beta}\partial_\mu, \tag{1.8}$$

$$\{Q_\alpha, Q_\beta\} = 0, \qquad \{Q^\dagger_\alpha, Q^\dagger_\beta\} = 0, \tag{1.9}$$

$$[Q_\alpha, P_\mu] = 0, \qquad [Q^\dagger_\alpha, P_\mu] = 0, \qquad [P_\mu, P_\nu] = 0. \tag{1.10}$$

The precise ordering of spinor indices is not stable in the lecture itself; Susskind discusses and revises it on the board. We therefore keep the index structure schematic where that is enough to express the logic.

The supercharges act as differential operators on superspace. In the lecture their detailed indices

are soon suppressed because, as he says, the indexology gets awful. In that same spirit we write

$$Q = \frac{\partial}{\partial\theta} + i\,\bar{\theta}\sigma^\mu\partial_\mu, \qquad \bar{Q} = \frac{\partial}{\partial\bar{\theta}} + i\,\sigma^\mu\theta\,\partial_\mu, \tag{1.11}$$

again with spinor contractions understood schematically. These operators generate supersymmetry transformations:

$$\delta\Phi = (\epsilon Q + \bar{\epsilon}\,\bar{Q})\Phi. \tag{1.12}$$

The lecture's point is simple and geometric. Just as $P_\mu$ generates infinitesimal shifts in $x^\mu$, the supercharges generate infinitesimal shifts in the Grassmann directions, accompanied by the compensating motion in space-time demanded by the algebra.

## 1.4 Chiral Superfields as a Consistent Constraint

At this stage the lecture changes register. It does not yet want component transformation laws. It wants manageable superfields. A completely general superfield depending on all of $x, \theta, \bar{\theta}$ is too elaborate for the present purpose. So the local problem becomes: can we impose a simplifying constraint and still remain inside a supersymmetric class of fields?

Susskind motivates this through ordinary symmetry. The condition

$$v_x = 0$$

is not rotationally invariant: rotate the vector and $v_x$ generally reappears. By contrast,

$$v^2 = 1$$

is preserved by rotations. The lesson is that a constraint must be tested against the symmetry. If the symmetry maps the constrained configurations back into constrained configurations, then the constraint is compatible.

### 1.4.1 Question & Answer

**Question.** What makes the constraint $\bar{D}\Phi = 0$ compatible with supersymmetry?

**Answer.** Suppose $\Phi$ obeys $\bar{D}\Phi = 0$. After a supersymmetry variation we must still have $\bar{D}(\Phi + \delta\Phi) = 0$. A sufficient condition is that the constraint operator commute or anticommute with the symmetry generator. Then

$$\bar{D}(Q\Phi) = \pm Q(\bar{D}\Phi) = 0,$$

and similarly for $\bar{Q}$. The transformed field stays within the constrained class. That is why the lecture first asks for operators that anticommute with the supercharges before imposing any specific constraint.

The operators used for this purpose are $D$ and $\bar{D}$, obtained from the supercharges by flipping the sign of the space-time shift term:

$$D = \frac{\partial}{\partial\theta} - i\,\bar{\theta}\sigma^\mu\partial_\mu, \qquad \bar{D} = \frac{\partial}{\partial\bar{\theta}} - i\,\sigma^\mu\theta\,\partial_\mu. \tag{1.13}$$

These are the differential operators from which supersymmetry-compatible constraints are built.

The lecture then chooses the special condition

$$\bar{D}\Phi = 0. \tag{1.14}$$

It is important to keep the logic straight. A chiral superfield is not introduced as a standard textbook object that one is simply expected to know. It emerges as the answer to the question, "What kind of simplifying constraint can we impose without breaking supersymmetry?"

## 1.5 Solving $\bar{D}\Phi = 0$ and Expanding the Chiral Superfield

We now solve the constraint. The lecture first compares it with the simpler equation $\partial\Phi/\partial\bar{\theta} = 0$. That equation would merely say that $\Phi$ does not depend on $\bar{\theta}$ at all. But $\bar{D}\Phi = 0$ is not that. It says that $\bar{\theta}$ may appear, but only through a very special combination with $x$.

Define

$$y^\mu = x^\mu + i\,\bar{\theta}\sigma^\mu\theta. \tag{1.15}$$

The claim is that the general solution of the chiral constraint is

$$\Phi = \Phi(y, \theta). \tag{1.16}$$

In other words, there is no independent $\bar{\theta}$-dependence; the only $\bar{\theta}$-dependence is hidden inside $y^\mu$.

The lecture checks this by the chain rule. If $\Phi$ depends on $\bar{\theta}$ only through $y^\mu$, then

$$\frac{\partial}{\partial\bar{\theta}_\alpha}\Phi(y,\theta) = \frac{\partial\Phi}{\partial y^\mu}\,\frac{\partial y^\mu}{\partial\bar{\theta}_\alpha} = i(\sigma^\mu\theta)_\alpha\,\partial_\mu\Phi. \tag{1.17}$$

But the second term in $\bar{D}_\alpha$ contributes precisely the opposite expression. Therefore

$$\bar{D}_\alpha\Phi = \left(\frac{\partial}{\partial\bar{\theta}_\alpha} - i(\sigma^\mu\theta)_\alpha\partial_\mu\right)\Phi = 0. \tag{1.18}$$

That is the whole mechanism. The $\bar{\theta}$-variation produced by the dependence of $y^\mu$ cancels the explicit space-time derivative term in $\bar{D}$.

This is the point where the lecture pauses and gives the superfield a name. A field obeying $\bar{D}\Phi = 0$ is a *chiral superfield*. It is also, in general, complex.

Since the chiral superfield depends only on $y^\mu$ and the two components of $\theta$, its Grassmann expansion terminates at order $\theta^2$:

$$\Phi(y,\theta) = \phi(y) + \theta\psi(y) + \theta^2 F(y). \tag{1.19}$$

We keep the notation deliberately schematic, as the lecture does. The three component fields are a complex scalar $\phi$, a two-component fermion $\psi$, and an auxiliary complex scalar $F$. Susskind already signals that $F$ is mathematically necessary but dynamically peculiar. That warning will matter as soon as the Lagrangian is unpacked.

## 1.6 Superspace Actions and the Free Example $\Phi^*\Phi$

The lecture now turns from kinematics to construction. A super-Lagrangian depends on $x, \theta, \bar{\theta}$, and the action is obtained by integrating over the whole superspace:

$$S = \int d^4x\, d^2\theta\, d^2\bar{\theta}\, \mathcal{L}_{\text{sup}}. \tag{1.20}$$

The Grassmann integrals simply pick out the highest component, that is, the coefficient of $\theta^2\bar{\theta}^2$. So the whole method reduces to a peculiar but efficient piece of bookkeeping: write down a superspace expression, expand it, and keep the top Grassmann term.

The first example is

$$\mathcal{L}_{\text{sup}} = \Phi^*\Phi. \tag{1.21}$$

That looks almost embarrassingly simple. There are no derivatives of the superfield anywhere. The lecture makes a point of this, because the result will nonetheless contain ordinary kinetic terms.

For the conjugate chiral field,

$$\Phi^* = \Phi^*(x^\mu - i\,\bar{\theta}\sigma^\mu\theta, \bar{\theta}). \tag{1.22}$$

Thus the action contains

$$\Phi^*(x - i\bar{\theta}\sigma\theta, \bar{\theta})\, \Phi(x + i\bar{\theta}\sigma\theta, \theta). \tag{1.23}$$

Before expanding, Susskind uses a trick that is both simple and important. Since we are integrating over $x$, we may shift the integration variable without changing the integral. If we shift $x$ so as to remove the Grassmann correction from one factor, it doubles in the other:

$$\Phi^*(x - i\bar{\theta}\sigma\theta, \bar{\theta})\, \Phi(x + i\bar{\theta}\sigma\theta, \theta) \longrightarrow \Phi^*(x, \bar{\theta})\, \Phi(x + 2i\bar{\theta}\sigma\theta, \theta). \tag{1.24}$$

The lecture lingers here because this shift is where the derivatives will come from. Without it, the super-Lagrangian $\Phi^*\Phi$ would never produce space-time propagation.

We now Taylor-expand the shifted argument. For the scalar component,

$$\phi(x + 2i\bar{\theta}\sigma\theta) = \phi(x) + 2i(\bar{\theta}\sigma^\mu\theta)\, \partial_\mu\phi(x) + (\bar{\theta}\sigma^\mu\theta)(\bar{\theta}\sigma^\nu\theta)\, \partial_\mu\partial_\nu\phi(x), \tag{1.25}$$

where, just as in the lecture, we suppress numerical coefficients beyond the clearly secure structure. The spinor term is treated similarly, but it truncates earlier because once too many $\theta$'s appear the expression vanishes. For the auxiliary term one need not expand the argument at all, since any additional shift would create too many Grassmann factors.

What survives the $d^2\theta\, d^2\bar{\theta}$ integration is therefore the $\theta^2\bar{\theta}^2$ coefficient. The board and transcript securely fix only the schematic structure, not the precise prefactors. In that schematic form the ordinary component Lagrangian is

$$\mathcal{L} \sim \phi^*\Box\phi + \bar{\psi}\,\sigma^\mu\partial_\mu\psi + F^*F. \tag{1.26}$$

Up to coefficients and integrations by parts, this is exactly what the blackboard is gathering on its lower line. The scalar term is the Klein–Gordon kinetic term, the fermionic term is the chiral Dirac kinetic term, and the last term is the algebraic $F$-term.

The lecture immediately draws the moral. The ordinary derivatives were not inserted by hand into the super-Lagrangian. They were generated by the shifted dependence on $x \pm i\bar{\theta}\sigma\theta$. Superspace looks abstract, but it is not inert. It turns a formally simple expression into a nontrivial space-time field theory.

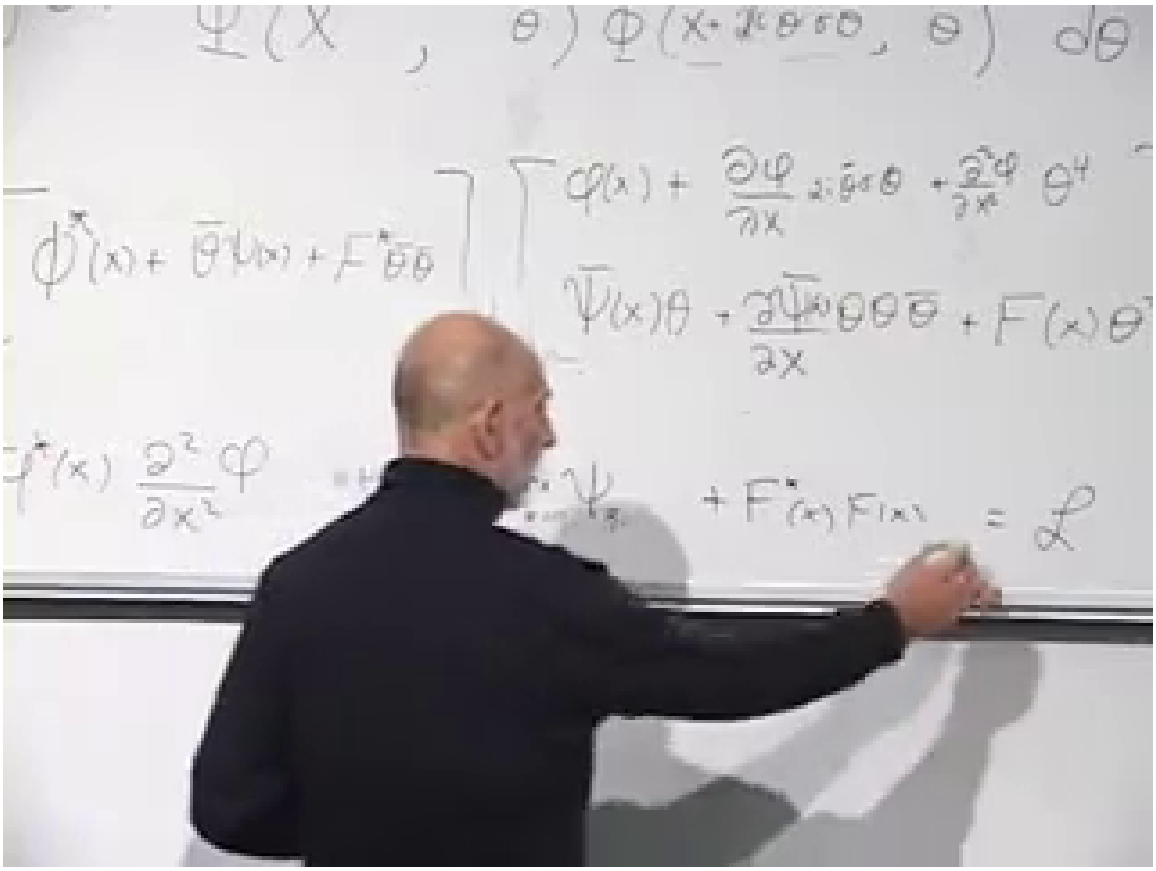

Figure 1.2: Component expansion leading to the ordinary Lagrangian.

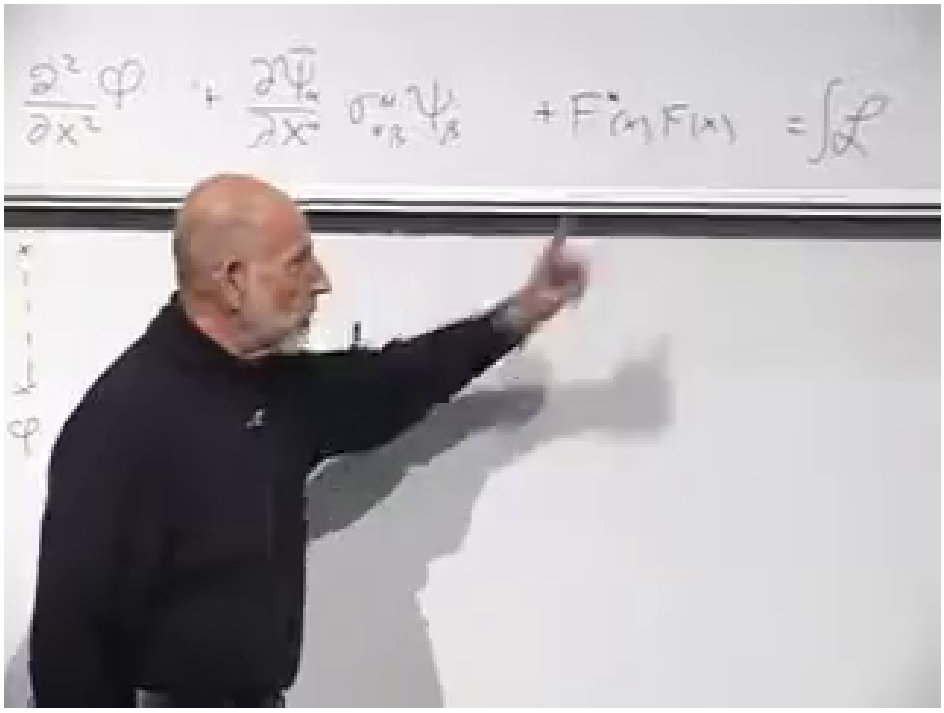

Figure 1.3: Auxiliary-field term in the simple supersymmetric Lagrangian.

## 1.7 Mass Terms, Interaction Terms, and the Role of the Auxiliary Field

Having obtained the free theory, the lecture asks for something more interesting. First come masses, then interactions. The guiding idea is the same throughout: write a very simple superspace expression, extract the relevant Grassmann coefficient, and let the auxiliary field translate part of the holomorphic structure into ordinary bosonic terms.

Before doing that, Susskind isolates the $F$-term once more. In the free theory we have

$$\mathcal{L} \supset F^* F. \tag{1.27}$$

The crop in Figure 1.3 is valuable precisely because it shows this piece at the place where the blackboard turns into the ordinary Lagrangian.

This term contains no derivatives. Therefore the Euler–Lagrange equation for $F$ is purely algebraic. In the free case,

$$\frac{\partial \mathcal{L}}{\partial F} = 0 \quad \Rightarrow \quad F^* = 0, \qquad \frac{\partial \mathcal{L}}{\partial F^*} = 0 \quad \Rightarrow \quad F = 0. \tag{1.28}$$

This is why the lecture says that $F$ is trivial. It is not a propagating boson in the ordinary sense.

### 1.7.1 Question & Answer

**Question.** Why is it legitimate to integrate only over $d^2\theta$ when the expression has no $\bar{\theta}$-dependence?

**Answer.** If one kept the full measure $d^2\theta\, d^2\bar{\theta}$, then a purely chiral expression would indeed vanish upon $\bar{\theta}$-integration. The lecture explicitly notices this and then uses the standard supersymmetric repair: when the integrand is chiral, one integrates only over the chiral measure $d^2\theta$. Susskind remarks that this deserves justification. At this point in the lecture, however, it is taken as the correct supersymmetric rule and then immediately used.

With that understood, consider the quadratic chiral term

$$S_m = \int d^4x\, d^2\theta\; m\, \Phi^2 + \text{c.c.} \tag{1.29}$$

Using the chiral expansion

$$\Phi = \phi + \theta\psi + \theta^2 F, \tag{1.30}$$

we extract the $\theta^2$ coefficient of $\Phi^2$. The lecture does this schematically rather than with exact combinatorial factors:

$$\Phi^2\big|_{\theta^2} \sim F\phi + \psi\psi. \tag{1.31}$$

Hence

$$\mathcal{L}_m \sim m\, F\phi + m\, \psi\psi + \text{c.c.} \tag{1.32}$$

The fermionic term is a two-component mass term; Susskind refers to it as a Majorana-type mass term.

The bosonic mass is hidden inside the auxiliary field. Combining the new term with the free $F^*F$ piece gives

$$\mathcal{L}(F) = F^*F + m\, F\phi + m\, F^*\phi^*. \tag{1.33}$$

Its equations of motion are

$$F^* + m\phi = 0, \qquad F + m\phi^* = 0. \tag{1.34}$$

Substituting back yields

$$\mathcal{L}_{\text{boson mass}} \sim m^2\, \phi^*\phi. \tag{1.35}$$

This is the first strong supersymmetric payoff of the lecture. The same supersymmetric structure that gives a fermion mass also gives the boson a matching mass. The notation looks different — $m$ in the fermionic equation, $m^2$ in the bosonic Lagrangian — but the physical mass is the same.

The next example is the cubic term

$$S_g = \int d^4x\, d^2\theta\; g\, \Phi^3 + \text{c.c.} \tag{1.36}$$

Again we keep only the $\theta^2$ part. The lecture identifies two schematic contributions:

$$\Phi^3\big|_{\theta^2} \sim \phi\, \psi\psi + \phi^2 F. \tag{1.37}$$

So the interaction Lagrangian contains

$$\mathcal{L}_{\text{int}} \sim g\, \phi\, \psi\psi + g\, \phi^*\bar{\psi}\bar{\psi} + g\, \phi^2 F + g\, \phi^{*2} F^*. \tag{1.38}$$

The first pair of terms is the Yukawa interaction. The second pair couples scalars to the auxiliary field. Because $F$ is not an ordinary propagating particle, these terms do not give a new long-range mode. Instead, eliminating $F$ generates an effective quartic bosonic interaction. In schematic form,

$$\mathcal{L}_{\text{int}} \sim g\, \phi\, \psi\psi + g\, \phi^*\bar{\psi}\bar{\psi} + g^2\, \phi^2\phi^{*2}. \tag{1.39}$$

The point is not the exact coefficient. The point is the relation: the four-boson coupling is tied to the square of the Yukawa coupling.

Figure 1.4: Pedagogical redraw of the $F$-field propagator as a local contact. The lecture's claim is that with no derivatives the propagation collapses to a delta function at one point.

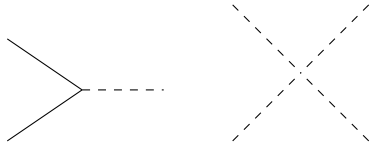

Figure 1.5: Pedagogical redraws of the interactions generated by the $\Phi^3$ term: the Yukawa vertex and the effective four-boson contact induced by the auxiliary field.

## 1.8 Feynman-Diagram Interpretation and the Supersymmetric Payoff

The lecture now shifts viewpoint once more. Instead of solving equations of motion, it asks what the component terms mean diagrammatically.

First, the auxiliary field. Since $F^*F$ contains no derivatives, the corresponding propagator does not spread from one point to another in the ordinary way. Susskind says that it is effectively a delta function: the line begins and ends at the same point. The little dashed doodle on the blackboard near Figure 1.3 supports exactly this reading, even though it is not a precise finished diagram. A clean pedagogical redraw is therefore useful.

Second, the cubic chiral term gives two visually different interactions that are numerically linked: the Yukawa vertex $\phi\psi\psi$ and the effective four-boson contact obtained after eliminating $F$.

This numerical linkage is the real content of supersymmetry here. One could always invent a boson and a fermion and simply assign them the same mass. That would be a free theory and nothing more. What is nontrivial is that once interactions are present, supersymmetry keeps relating apparently different couplings.

The lecture's final concrete example is the boson self-energy. There are two lowest-order contributions worth comparing. One comes from the bosonic quartic interaction. The other comes from a fermion loop built from two Yukawa vertices. The lecture does not evaluate the integrals in detail. It only asks for their structure and their coefficients. The first comes with a coefficient of order $g^2$. The second also comes with a coefficient of order $g^2$, since there is one $g$ at each Yukawa vertex. The masses running in the loops are also matched by supersymmetry.

Now the payoff can be stated clearly. Supersymmetry does not merely place a boson and a fermion in the same multiplet. It enforces equal masses and specific relations among couplings, and those relations are precisely what allow bosonic and fermionic loop effects to cancel. The lecture stresses

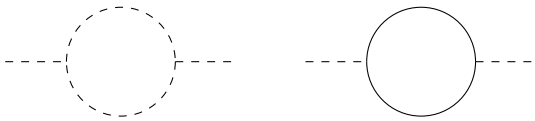

Figure 1.6: Schematic paired boson self-energy diagrams. The left loop is bosonic; the right loop is fermionic. Supersymmetry ties the masses and couplings together, while fermion and boson loops contribute with opposite signs.

that this pairing persists to all orders. That is the real force of the construction.

Only after making that point does Susskind step back and remind us that exact supersymmetry is not what nature shows us directly. So the lecture ends by pointing forward: one must next understand supersymmetry breaking, and then one may ask how the whole structure influences grand unification, where $SU(3) \times SU(2) \times U(1)$ is invited to enlarge to something like $SU(5)$.

## 1.9 Summary

The lecture begins from a defect in the time-only toy model and uses that defect to drive the whole discussion forward. If $\{Q^\dagger, Q\}$ closes only on the Hamiltonian, then Lorentz invariance is already in danger. That difficulty leads us first to the massless chiral fermion and its Hamiltonian $H = \sigma \cdot p$, then to the covariant chiral equation $i\sigma^\mu \partial_\mu \psi = 0$, and only then to spinorial Grassmann coordinates and the four-dimensional superalgebra.

Once superspace is in place, the lecture turns to the question of simplification. The chiral constraint $\bar{D}\Phi = 0$ is introduced not as a standard definition but as a consistency-preserving reduction of the most general superfield. Solving that constraint gives the chiral superfield $\Phi(y, \theta)$, and the superspace action principle then unfolds into an ordinary field theory. The free example $\Phi^*\Phi$ produces scalar and fermion kinetic terms together with the auxiliary-field term $F^*F$. The quadratic term $\Phi^2$ gives equal boson and fermion masses. The cubic term $\Phi^3$ gives a Yukawa coupling and an induced quartic bosonic interaction whose coefficient is fixed by supersymmetry. Finally, the lecture interprets these relations diagrammatically: equal masses and linked couplings are what make the bosonic and fermionic loop contributions cancel.

That is the proper closing note. Exact supersymmetry is extraordinarily rigid and powerful. It is not manifest as an exact symmetry of nature, but precisely for that reason the next problems are compelling: how supersymmetry breaks, and how its presence may reshape the unification of forces.